

5 DTrace

According to many, what makes a programmer is his/her ability to write optimised code that goes on to build better performing applications. This involves keeping many aspects of programming like memory consumption and I/O operations in mind while developing an application. While it is possible to achieve one level of optimization by skillfully writing code, but it becomes increasingly difficult as the size and the complexity of the application increases. Therefore developers often feel a need for a tool which could explain them how exactly their application is behaving while it is under execution.

There is a second aspect to consider as well. Imagine that you have a web application which has thousands of users connected to it, (say something like Gmail or Facebook). Now, if a bug is discovered in your application, you cannot play around with it while the application is running on the deployment machines (servers with which other users are connected) because of performance as well as security reasons. Most of the time, the bug is recreated in a development environment on to which the developers try to find and fix the problem, but it is a very time consuming and tedious process. Therefore, both system administrators as well as software developers feel a need for a tool, which can help them find out faults in their application, while the application is still running, without hampering the performance and compromising with security.

DTrace is an answer to all these questions. Its a tool which software developers can use to optimize and debug their programs and also use to understand the behavior of their application, while the system administrators can use it for finding problem points in an application while it is still running in production environments. In this chapter, we will like to bring out a detailed overview of DTrace. We will use the DTrace Quick Start Guide published by Sun Microsystems Inc. for this purpose. The complete guide can be downloaded from http://blogs.sun.com/observatory/entry/dtrace_quick_start_guide and it is also included in the book. For a deeper understanding of DTrace, we recommend that you read Solaris Dynamic Tracing Guide, which is in the Digit

Quick Tip

Before you begin with this chapter, we recommend that you get familiar with Unix Shell Scripting. You can learn basics of shell programming at <http://tr.im/unixshell> If you are already familiar with shell scripting then go ahead and enjoy the chapter!